For the use only of a Registered Medical Practitioner

PRESCRIBING INFORMATION

RANOZYP TABLETS

(Olanzapine Tablets 5/10 mg)

COMPOSITION

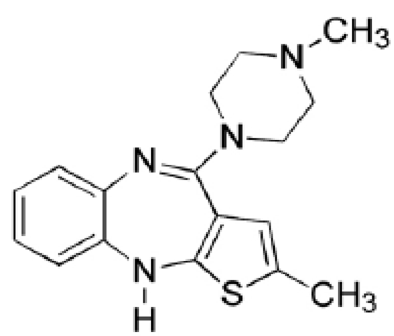
Ranozyp Tablets 5 mg
Each tablet contains:
Olanzapine…………………..5 mg

Ranozyp Tablets 10 mg
Each tablet contains:
Olanzapine………………….. 10 mg

Excipients:
Lactose anhydrous
Microcrystalline Cellulose
L-Hydroxy propyl cellulose
Magnesium Stearate

DESCRIPTION

RANOZYP TABLETS contain olanzapine, is an atypical psychotropic that belongs to the thienobenzodiazepine class. It is chemically designated as 2-methyl-4-(4-methyl-1-piperazinyl)-10*H*-thieno[2,3-*b*] [1,5]benzodiazepine. The molecular formula is $C_{17}H_{20}N_4S$, which corresponds to a molecular weight of 312.44. The chemical structure is:



Olanzapine

Product Description:

For 5 mg strength: Light yellow to yellow coloured, round, biconvex tablets debossed with ‘O5’ on one side and plain on other side.

For 10 mg strength: Light yellow to yellow coloured, round, biconvex tablets debossed with ‘O7’ on one side and plain on other side.

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

- Pharmacodynamics

Olanzapine binds with high affinity to the following receptors: serotonin $5HT_{2A/2C}$, $5HT_6$(Ki=4, 11, and 5 nM, respectively), dopamine D_{1-4} (Ki=11-31 nM), histamine H_1(Ki=7 nM), and adrenergic $\alpha_1$receptors (Ki=19 nM). Olanzapine is an antagonist with moderate affinity binding for serotonin $5HT_3$(Ki=57 nM) and muscarinic M1-5 (Ki=73, 96, 132, 32, and 48 nM, respectively). Olanzapine binds weakly to GABAA, BZD, and β-adrenergic receptors (Ki>10 μM).

Antagonism at receptors other than dopamine and $5HT_2$ may explain some of the other therapeutic and side effects of olanzapine. Olanzapine’s antagonism of muscarinic M_{1-5} receptors may explain its anticholinergic-like effects. Olanzapine’s antagonism of histamine H_1 receptors may explain the somnolence observed with this drug. Olanzapine’s antagonism of adrenergic $\alpha_1$receptors may explain the orthostatic hypotension observed with this drug.

- Pharmacokinetics

Olanzapine is well absorbed after oral administration, reaching peak plasma concentrations within 5 to 8 hours. The absorption is not affected by food.

Olanzapine is metabolized in the liver by conjugative and oxidative pathways. The major circulating metabolite is the 10-N-glucuronide, which does not pass the blood brain barrier. Cytochromes P450-CYP1A2 and P450-CYP2D6 contribute to the formation of the N-desmethyl and 2-hydroxymethyl metabolites; both exhibited significantly less *in vivo* pharmacological activity than olanzapine in animals. The predominant pharmacologic activity is from the parent, olanzapine. After oral administration, the mean terminal elimination half-life of olanzapine varies on the basis of age and gender.

The mean elimination half-life in elderly (65 and over) is reported to be prolonged and the clearance is reduced as compared to non elderly. The pharmacokinetic variability in the elderly is within the range for the non-elderly. In patients with schizophrenia > 65 years of age, dosing from 5 to 20 mg/day is not associated with any distinguishing profile of adverse events.

The mean elimination half-life in females is prolonged and the clearance reduced as compared to males. However, safety profile of olanzapine (5-20 mg) is comparable between male and female patients.

No significant difference has been reported in mean elimination half-life or clearance in renally impaired patients (creatinine clearance <10 ml/min). Approximately 57% of radiolabeled olanzapine appeared in urine, principally as metabolites.

In smoking subjects with mild hepatic dysfunction, mean elimination half-life has been reported to be prolonged and clearance reduced.

In non-smokers, prolongation of mean elimination half-life and reduction in clearance has been reported.

The plasma clearance of olanzapine is reported to be lower in elderly, in females, and in non-smokers. However, the magnitude of the impact of age, gender, or smoking on olanzapine clearance and half-life is reported to be small.

There is no difference in the pharmacokinetic parameters among the Caucasians, Japanese, and Chinese populations.

The plasma protein binding of olanzapine is about 93% over the concentration range of about 7 to about 1,000 ng/ml. Olanzapine is bound predominantly to albumin and α_1-acid-glycoprotein.

Paediatric population

Adolescents (ages 13 to 17 years): The pharmacokinetics of olanzapine is reported to be similar between adolescents and adults. The average olanzapine exposure was reported to be higher in adolescents which could be due to demographic differences between the adolescents and adults.

INDICATIONS

Olanzapine is indicated for the treatment of schizophrenia.

Olanzapine is effective in maintaining the clinical improvement during continuing therapy in patients who have shown initial treatment response.

Olanzapine is indicated for short term treatment of acute manic episodes associated with Bipolar I Disorder.

Olanzapine is indicated for preventing recurrence of manic, mixed or depressive episodes in Bipolar I Disorder.

DOSE AND METHOD OF ADMINISTRATION

Schizophrenia
The recommended starting dose for olanzapine is 10 mg/day.

Manic episode:
The starting dose is 15 mg as a single daily dose in monotherapy or 10 mg daily in combination therapy.

Preventing recurrence in bipolar disorder:
The recommended starting dose is 10mg/day. For patients who have been receiving olanzapine for treatment of manic episode, continue therapy for preventing recurrence at the same dose. If a new manic, mixed, or depressive episode occurs, olanzapine treatment should be continued

(with dose optimization as needed), with supplementary therapy to treat mood symptoms, as clinically indicated.

During treatment for both schizophrenia, manic episode and recurrence prevention in bipolar disorder, daily dosage may subsequently be adjusted on the basis of individual clinical status within the range 5-20 mg/day. An increase to a dose greater than the recommended starting dose is advised only after appropriate clinical reassessment and should generally occur at intervals of not less than 24 hours. Olanzapine can be given without regards for meals as absorption is not affected by food. Gradual tapering of the dose should be considered when discontinuing olanzapine.

Children:
Olanzapine is not recommended for use in children and adolescent below 18 years of age due to lack of data on safety and efficacy.

Elderly patients:
A lower starting dose (5 mg/day) is not routinely indicated but should be considered for those 65 and over when clinical factors warrant.

Patients with renal and/or hepatic impairment:
A lower starting dose (5 mg) should be considered for such patients. In cases of moderate hepatic insufficiency (cirrhosis, Child-Pugh Class A or B), the starting dose should be 5 mg and only increased with caution.

Gender:
The starting dose and dose range need not be routinely altered for female patients relative to male patients.

Smokers:
The starting dose and dose range need not be routinely altered for non-smoking patients relative to smoking patients.

When more than one factor is present which might result in slower metabolism (female gender, geriatric age, non-smoking status), consideration should be given to decreasing the starting dose. Dose escalation: when indicated, should be conservative in such patients.

CONTRAINDICATIONS

RANOZYP TABLETS (Olanzapine) is contraindicated

- *in patients with a known hypersensitivity to olanzapine or any of the excipients.*
- *in patients with known risk for narrow-angle glaucoma.*

WARNINGS AND PRECAUTIONS

WARNINGS

During antipsychotic treatment, improvement in the patient's clinical condition may take several days to some weeks. Patients should be closely monitored during this period.

Dementia-related psychosis and/or behavioral disturbances

Olanzapine is not approved for the treatment of dementia-related psychosis and/or behavioral disturbances and is not recommended for use in this particular group of patients because of an increase in mortality and the risk of cerebrovascular accident. In elderly patients (mean age 78 years) with dementia-related psychosis and/or disturbed behaviors, there was a 2 fold increase in the incidence of death in olanzapine treated patients. The higher incidence of death was not associated with olanzapine dose (mean daily dose 4.4 mg) or duration of treatment. Risk factors that may predispose this patient population to increased mortality include age > 65 years, dysphagia, sedation, malnutrition and dehydration, pulmonary conditions (e.g., pneumonia, with or without aspiration), or concomitant use of benzodiazepines. However, the incidence of death was higher in olanzapine-treated patients independent of these risk factors.

Cerebrovascular adverse events (CVAE e.g., stroke, transient ischaemic attack), including fatalities, have been reported in elderly patients with dementia-related psychosis. There was a 3-fold increase in CVAE in patients treated with olanzapine. All patients who experienced a cerebrovascular event had pre-existing risk factors. Age > 75 years and vascular/mixed type dementia were identified as risk factors for CVAE in association with olanzapine treatment. The efficacy of olanzapine in treatment of dementia-related psychosis and/or behavioral disturbances is not established.

Parkinson's disease

The use of olanzapine in the treatment of dopamine agonist associated psychosis in patients with Parkinson's disease is not recommended. Worsening of Parkinsonian symptomatology and hallucinations has been reported very commonly and olanzapinewas not reported to be effective in the treatment of psychotic symptoms. Patients were initially required to be stable on the lowest effective dose of anti-Parkinsonian medicinal products (dopamine agonist) and to remain on the same anti-Parkinsonian medicinal products and dosages. Olanzapine was started at the starting dose of 2.5 mg/day and titrated to a maximum of 15 mg/day.

Suicide

The possibility of a suicide attempt is inherent in schizophrenia and in bipolar I disorder, and close supervision of high-risk patients should accompany drug therapy. Prescriptions for olanzapine should be written for the smallest quantity of tablets consistent with good patient management, in order to reduce the risk of overdose.

Neuroleptic Malignant Syndrome (NMS)

NMS is a potentially life-threatening condition associated with antipsychotic medicinal product. Rare cases reported as NMS have also been reported in association with olanzapine. Clinical manifestations of NMS are hyperpyrexia, muscle rigidity, altered mental status, and evidence of autonomic instability (irregular pulse or blood pressure, tachycardia, diaphoresis, and cardiac dysrhythmia). Additional signs may include elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure. If a patient develops signs and symptoms indicative of NMS, or presents with unexplained high fever without additional clinical manifestations of NMS, all antipsychotic medicines, including olanzapine must be discontinued.

Hyperglycemia and Diabetes Mellitus

Hyperglycemia in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients treated with atypical antipsychotics. Assessment of the relationship between atypical antipsychotics use and glucose abnormalities is complicated by the possibility of an increased background risk of diabetes mellitus in patients with schizophrenia and the increasing incidence of diabetes mellitus in the general population. Given this confounders, the relationship between atypical antipsychotic use and hyperglycemia-related adverse events is not completely understood. However, reported epidemiological data suggest an increased risk of treatment-emergent hyperglycemia-related events in patients treated with the atypical antipsychotics.

Precise risk estimates for hyperglycemia-related adverse events in patients treated with atypical antipsychotics are not available. Patients with an established diagnosis of diabetes mellitus who are started on atypical antipsychotics should be monitored regularly for worsening of glucose control. Patients with risk factors for diabetes mellitus (e.g. obesity, family history of diabetes) who are starting treatment with atypical antipsychotics should undergo fasting blood glucose testing at the beginning of treatment and periodically during treatment. Any patient treated with atypical antipsychotics should be monitored for symptoms of hyperglycemia including polydipsia, polyuria, polyphagia, and weakness. Patients who develop symptoms of hyperglycemia during treatment with atypical antipsychotics should undergo fasting blood glucose testing. In some cases, hyperglycemia has resolved when the atypical antipsychotic was discontinued; however, some patients required continuation of anti-diabetic treatment despite discontinuation of the suspect drug.

During antipsychotic treatment, improvement in the patient's clinical condition may take several days to some weeks. Patients should be closely monitored during this period.

Lipid alterations

Undesirable alterations in lipids have been reported in olanzapine-treated patients. Lipid alterations should be managed as clinically appropriate, particularly in dyslipidemic patients and in patients with risk factors for the development of lipids disorders. Patients treated with any antipsychotic agents, including olanzapine, should be monitored regularly for lipids in accordance with utilized antipsychotic guidelines, e.g., at baseline, 12 weeks after starting olanzapine treatment and every 5 years thereafter.

Anticholinergic activity

While olanzapine demonstrated anticholinergic activity *in vitro*, a low incidence of related events have been reported. However, as experience with olanzapine in patients with concomitant illness is limited, caution is advised when prescribing for patients with prostatic hypertrophy, or paralytic ileus and related conditions.

Hepatic function

Transient, asymptomatic elevations of hepatic aminotransferases, alanine transferase (ALT), aspartate transferase (AST) have been reported commonly, especially in early treatment. Caution should be exercised and follow-up organized in patients with elevated ALT and/or AST, in patients with signs and symptoms of hepatic impairment, in patients with pre-existing conditions associated with limited hepatic functional reserve, and in patients who are being treated with potentially hepatotoxic medicines. In the event of elevated ALT and/or AST during treatment, follow-up should be organised and dose reduction should be considered. In cases where hepatitis (including hepatocellular, cholestatic or mixed liver injury) has been diagnosed, olanzapine treatment should be discontinued.

Neutropenia
Caution should be exercised in patients with low leucocyte and/or neutrophil counts for any reason, in patients receiving medicines known to cause neutropenia, in patients with a history of drug-induced bone marrow depression/toxicity, in patients with bone marrow depression caused by concomitant illness, radiation therapy or chemotherapy and in patients with hypereosinophilic conditions or with myeloproliferative disease. Neutropenia has been reported commonly when olanzapine and valproate are used concomitantly.

Discontinuation of treatment
Acute symptoms such as sweating, insomnia, tremor, anxiety, nausea, or vomiting have been reported very rarely when olanzapine is stopped abruptly.

QT interval
Clinically meaningful QTc prolongations (Fridericia QT correction [QTcF] ≥ 500 milliseconds [msec] at any time post-baseline in patients with baseline QTcF < 500 msec) were uncommon in patients treated with olanzapine, with no significant differences in associated cardiac events. However, as with other antipsychotics, caution should be exercised when olanzapine is prescribed with medicines known to increase QTc interval, especially in the elderly, in patients with congenital long QT syndrome, congestive heart failure, heart hypertrophy, hypokalaemia or hypomagnesaemia.

Thromboembolism
Temporal association of olanzapine treatment and venous thromboembolism has very rarely been reported. A causal relationship between the occurrence of venous thromboembolism and treatment with olanzapine has not been established. However, since patients with schizophrenia often present with acquired risk factors for venous thromboembolism, all possible risk factors of VTE e.g., immobilisation of patients should be identified and preventive measures undertaken.

General CNS activity
Given the primary CNS effects of olanzapine, caution should be used when it is taken in combination with other centrally acting medicines and alcohol. As it exhibits *in vitro* dopamine antagonism, olanzapine may antagonise the effects of direct and indirect dopamine agonists.

Seizures
Olanzapine should be used cautiously in patients who have a history of seizures or are subject to factors which may lower the seizure threshold. Seizures have been reported to occur rarely in patients when treated with olanzapine. In most of these cases, a history of seizures or risk factors for seizures was reported.

Tardive dyskinesia
Olanzapine was reported to be associated with a statistically significant lower incidence of treatment-emergent dyskinesia. However, the risk of tardive dyskinesia increases with long-term exposure, and therefore if signs or symptoms of tardive dyskinesia appear in a patient on olanzapine, a dose reduction or discontinuation should be considered. These symptoms can temporally deteriorate or even arise after discontinuation of treatment.

Postural hypotension
Postural hypotension has been infrequently reported in the elderly patients treated with olanzapine. As with other antipsychotics, it is recommended that blood pressure is measured periodically in patients over 65 years.

Sudden cardiac death
In postmarketing reports with olanzapine, the event of sudden cardiac death has been reported in patients with olanzapine. The reported risk of presumed sudden cardiac death in patients treated with olanzapine was approximately twice the risk in patients not using antipsychotics. The risk of olanzapine was reported to be comparable to the risk of atypical antipsychotics included in a pooled analysis.

Paediatric population
Olanzapine is not indicated for use in the treatment of children and adolescents. Studies in patients aged 13-17 years reported various adverse reactions, including weight gain, changes in metabolic parameters and increases in prolactin levels. Long-term outcomes associated with these events have not been studied and remain unknown.

Lactose
This product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency, or glucose-galactose malabsorption should not take this medicine.

Weight Gain
Potential consequences of weight gain should be considered prior to starting olanzapine. Patients receiving olanzapine should receive regular monitoring of weight.

Dysphagia
Esophageal dysmotility and aspiration have been associated with antipsychotic drug use. Aspiration pneumonia is a common cause of morbidity and mortality in patients with advanced Alzheimer's disease. Olanzapine is not approved for the treatment of patients with Alzheimer's disease.

Body Temperature Regulation
Disruption of the body's ability to reduce core body temperature has been attributed to antipsychotic agents. Appropriate care is advised when prescribing olanzapine for patients who will be experiencing conditions which may contribute to an elevation in core body temperature, e.g., exercising strenuously, exposure to extreme heat, receiving concomitant medication with anticholinergic activity, or being subject to dehydration.

Use in Patients with Concomitant Illness
Clinical experience with olanzapine in patients with certain concomitant systemic illnesses is limited.

Olanzapine exhibits *in vitro* muscarinic receptor affinity. Olanzapine has been reported to be associated with constipation, dry mouth, and tachycardia, all adverse reactions possibly related to cholinergic antagonism. Olanzapine should be used with caution in patients with clinically significant prostatic hypertrophy, narrow angle glaucoma, or a history of paralytic ileus or related conditions.

The following treatment-emergent adverse reactions have been reported in olanzapine-treated elderly patients with dementia-related psychosis: falls, somnolence, peripheral edema, abnormal gait, urinary incontinence, lethargy, increased weight, asthenia, pyrexia, pneumonia, dry mouth and visual hallucinations. The rate of discontinuation due to adverse reactions has been reported with olanzapine. Elderly patients with dementia-related psychosis treated with olanzapine are at an increased risk of death. Olanzapine is not approved for the treatment of patients with dementia-related psychosis.

There is no information of olanzapine being evaluated or used to any appreciable extent in patients with a recent history of myocardial infarction or unstable heart disease. Because of the risk of orthostatic hypotension with olanzapine, caution should be observed in cardiac patients.

Hyperprolactinemia
As with other drugs that antagonize dopamine D_2 receptors, olanzapine elevates prolactin levels, and the elevation persists during chronic administration. Hyperprolactinemia may suppress hypothalamic GnRH, resulting in reduced pituitary gonadotropin secretion. This, in turn, may inhibit reproductive function by impairing gonadal steroidogenesis in both female and male patients. Galactorrhea, amenorrhea, gynecomastia, and impotence have been reported in patients receiving prolactin-elevating compounds. Long-standing hyperprolactinemia when associated with hypogonadism may lead to decreased bone density in both female and male subjects.

Tissue culture experiments reported indicate that approximately one-third of human breast cancers are prolactin dependent *in vitro*, a factor of potential importance if the prescription of these drugs is contemplated in a patient with previously detected breast cancer. As is common with compounds which increase prolactin release, an increase in mammary gland neoplasia was reported in the olanzapine carcinogenicity data in mice and rats. An association between chronic administration of this class of drugs and tumorigenesis in humans is not reported; the available reported evidence is considered too limited to be conclusive at this time.

Changes from normal to high in prolactin concentrations has been reported in adults treated with olanzapine. In adults treated with olanzapine, potentially associated clinical manifestations included menstrual-related event, sexual function-related events and breast-related events.

In adolescent patients with schizophrenia or bipolar I disorder (manic or mixed episodes) on olanzapine monotherapy, changes from normal to high in prolactin concentrations has been reported. In adolescents treated with olanzapine, potentially associated clinical manifestations included menstrual-related events, sexual function-related events, and breast-related events.

Laboratory Tests
Fasting blood glucose testing and lipid profile at the beginning of, and periodically during, treatment is recommended.

Effects on ability to drive and use machines
No studies on the effects on the ability to drive and use machines have been performed. Because olanzapine may cause somnolence and dizziness, patients should be cautioned about operating machinery, including motor vehicles.

Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)
Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) has been reported with olanzapine exposure. DRESS consists of a combination of three or more of the following: cutaneous reaction (such as rash or exfoliative dermatitis), eosinophilia, fever, lymphadenopathy and one or more systemic complications such as hepatitis, nephritis, pneumonitis, myocarditis, and pericarditis. Discontinue olanzapine if DRESS is suspected.

DRUG INTERACTIONS

Paediatric population: Reported interaction information is available only for adult population.

Potential Interactions Affecting Olanzapine
Since olanzapine is metabolized by CYP1A2, substances that can specifically induce or inhibit this isoenzyme may affect the pharmacokinetics of olanzapine.

Induction of CYP1A2: The metabolism of olanzapine may be induced by smoking and carbamazepine, which may lead to reduced olanzapine concentrations. Only slight to moderate increase in olanzapine clearance has been reported. The clinical consequences are likely to be limited, but clinical monitoring is recommended and an increase of olanzapine dose may be considered if necessary.

Inhibition of CYP1A2: Fluvoxamine, a specific CYP1A2 inhibitor, has been reported to significantly inhibit the metabolism of olanzapine. The mean increase in olanzapine C_{max} following fluvoxamine was reported to be 54% in female non-smokers and 77% in male smokers. The mean increase in olanzapine AUC was reported to be 52% and 108%, respectively. A lower starting dose of olanzapine should be considered in patients who are using fluvoxamine or any other CYP1A2 inhibitors, such as ciprofloxacin or ketoconazole. A decrease in the dose of olanzapine should be considered if treatment with an inhibitor of CYP1A2 is initiated.

Decreased bioavailability: Activated charcoal reduces the bioavailability of oral olanzapine by 50 to 60% and should be taken at least 2 hours before or after olanzapine.

Fluoxetine (a CYP2D6 inhibitor), single doses of antacid (aluminium, magnesium) or cimetidine have not been found to significantly affect the pharmacokinetics of olanzapine.

Alcohol: Ethanol (45 mg/70 kg single dose) did not have an effect on olanzapine pharmacokinetics. The co-administration of alcohol (i.e., ethanol) with olanzapine potentiated the orthostatic hypotension reported with olanzapine. Patients should be advised to avoid alcohol while taking olanzapine.

*Diazepam:*The co-administration of diazepam with olanzapine potentiated the orthostatic hypotension reported with olanzapine.

*Warfarin:*Warfarin (20 mg single dose) did not affect olanzapine pharmacokinetics.

*Inducers of CYP1A2 or Glucuronyl Transferase:*Omeprazole and rifampin may cause an increase in olanzapine clearance.

Potential for Olanzapine to Affect Other Medicinal Products

Olanzapine may antagonize the effects of direct and indirect dopamine agonists.

Olanzapine does not inhibit the main CYP450 isoenzymes *in vitro* (e.g., 1A2, 2D6, 2C9, 2C19, 3A4). Thus, no particular interaction is expected. No inhibition of metabolism of the following active substances was reported: tricyclic antidepressant (representing mostly CYP2D6 pathway), warfarin (CYP2C9), theophylline (CYP1A2), or diazepam (CYP3A4 and 2C19).

No interaction has been reported with olanzapine when co-administered with lithium or biperiden.

Therapeutic monitoring of valproate plasma levels did not indicate that valproate dosage adjustment is required after the introduction of concomitant olanzapine.

General CNS activity: Caution should be exercised in patients who consume alcohol or receive medicinal products that can cause central nervous system depression.

The concomitant use of olanzapine with anti-Parkinsonian medicinal products in patients with Parkinson's disease and dementia is not recommended.

QTc interval: Caution should be used if olanzapine is being administered concomitantly with medicinal products known to increase QTc interval.

*Antihypertensive Agents:*Olanzapine, because of its potential for inducing hypotension, may enhance the effects of certain antihypertensive agents.

SIDE EFFECTS/UNDESIRABLE EFFECTS

Adults

The most frequently reported adverse reactions associated with the use of olanzapine are somnolence, weight gain, eosinophilia, elevated prolactin, cholesterol, glucose and triglyceride levels, glucosuria, increased appetite, dizziness, akathisia, parkinsonism, dyskinesia, orthostatic hypotension, anticholinergic effects, transient asymptomatic elevations of hepatic transaminases, rash, asthenia, fatigue and oedema.
The following adverse reactions and laboratory investigations has been reported with olanzapine.

Blood and the lymphatic system disorders
Common: Eosinophilia
Uncommon:Leucopenia, neutropenia
Not known:Thrombocytopenia
Immune system disorders
Not known:Allergic reaction
Metabolism and nutrition disorders
Very Common:Weight gain
Common: Elevated cholesterol levels, elevated glucose levels, elevated triglyceride levels, glucosuria, increased appetite
Not known:Development or exacerbation of diabetes occasionally associated with ketoacidosis or coma, including some fatal cases, hypothermia

Nervous system disorders
Very Common: Somnolence

Common:Dizziness, akathisia, parkinsonism, dyskinesia
Not known: Seizures where in most cases a history of seizures or risk factors for seizures were reported, neuroleptic malignant syndrome, dystonia (including oculogyration), tardive dyskinesia, discontinuation symptoms

Cardiac disorders

Uncommon: Bradycardia, QTc prolongation
Not known: Ventricular tachycardia/fibrillation, sudden death
Vascular disorders
Common: Orthostatic hypotension
Not known: Thromboembolism (including pulmonary embolism and deep vein thrombosis)

Gastrointestinal disorders
Common: Mild, transient anticholinergic effects including constipation and dry mouth
Not known: Pancreatitis

Hepato-biliary disorders
Common:Transient, asymptomatic elevations of hepatic transaminases (ALT, AST), especially in early treatment.
Not known:Hepatitis (including hepatocellular, cholestatic or mixed liver injury)

Skin and subcutaneous tissue disorders
Common: Rash
Uncommon: Photosensitivity reaction, alopecia
Very rare: Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS).

Musculoskeletal and connective tissue disorders
Not known:Rhabdomyolysis

Renal and urinary disorders
Uncommon:Urinary incontinence
Not known:Urinary hesitation,

Reproductive system and breast disorders
Not known:Priapism

General disorders and administration site conditions
Common: Asthenia, fatigue, oedema

Investigations
Very common: Elevated plasma prolactin levels
Uncommon: High creatinine phosphokinase, increased total bilirubin
Not known: Increased alkaline phosphatase

Long-term exposure (at least 48 weeks)
Patients who had adverse, clinically significant changes in weight gain, glucose, total/LDL/HDL cholesterol or triglycerides increased over time. In adult patients on long-term therapy for 9-12 months, the rate of increase in mean blood glucose slowed after approximately 6 months.

Other reported adverse events are: Personality disorder, dyspepsia, accidental injury, back pain, chest pain, hypertension, eccymosis, extremity pain, joint pain, articulation impairment, rhinitis, cough increased, pharyngitis, amblyopia, increased salivation, amnesia, paresthesia, thirst, apathy, confusion, euphoria, in coordination, dysponea, acne, dry skin, abnormal vision, dysmenorrheal, vaginitis, chills, suicide attempt, hangover effect, vasodilation, tongue edema, ileus, intestinal obstruction, liver fatty deposit, hypoproteinemia, osteoporosis, ataxia, stupor, epistaxis, lung edema, abnormality of accommodation, dry eyes, mydriasis, amenorrhea, breast pain, decreased menstruation, impotence, menorrhagia, metrorrhagia, polyuria, urinary frequency, urinary retention, urination impaired, injection site pain, syncope, creatinine phosphokinase increased, sinusitis, arthralgia.

Additional information on special populations
In elderly patients with dementia, olanzapine treatment is associated with a higher incidence of death and cerebrovascular adverse reactions. Very common adverse reactions associated with the use of olanzapine in this patient group were abnormal gait and falls. Pneumonia, increased body temperature, lethargy, erythema, visual hallucinations and urinary incontinence were observed commonly.

In patients with drug-induced (dopamine agonist) psychosis associated with Parkinson's disease, worsening of Parkinsonian symptomatology and hallucinations were reported.

In patients with bipolar mania, valproate combination therapy with olanzapine neutropenia has been reported; a potential contributing factor could be high plasma valproate levels. Olanzapine administered with lithium or valproate resulted in increased levels of tremor, dry mouth, increased appetite, and weight gain. Speech disorder was also reported commonly. During treatment with olanzapine in combination with lithium or divalproex up to 6 weeks, an increase of baseline body weight has been reported. Also, long-term olanzapine treatment (up to 12 months) was associated with an increase from baseline body weight.

Paediatrics
Olanzapine is not indicated for the treatment of children and adolescent patients below 18 years.

The adverse reactions have been reported with a greater frequency in adolescent patients (aged 13-17 years). Clinically significant weight gain occurs more frequently in the adolescent population. The magnitude of weight gain is greater with long-term exposure (at least 24 weeks) than with short-term exposure.

Metabolism and nutrition disorders
*Very common:*Weight gain, elevated triglyceride levels, increased appetite.
*Common:*Elevated cholesterol levels.

Nervous system disorders
*Very common:*Sedation (including: hypersomnia, lethargy, somnolence).

Gastrointestinal disorders
*Common:*Dry mouth.

Hepato-biliary disorders

*Very common:*Elevations of hepatic transaminases (ALT/AST).

Investigations
*Very common:*Decreased total bilirubin, increased GGT, elevated plasma prolactin levels .

ECG Changes
In adults as well as adolescents, no significant differences are reported in ECG parameters, including QT, QTc (Fridericia corrected), and PR intervals. Olanzapine use is associated with a mean increase in heart rate.

USE IN SPECIAL POPULATION

- Pregnancy and Lactation

Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extra pyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor,somnolence, respiratory distress, and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalization.

RANOZYP TABLETS should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

Olanzapine is excreted in breast milk of healthy breast feeding women.. Patients should be advised not to breast feed an infant if they are taking olanzapine.

- Labor and Delivery

The effect of olanzapine on labor and delivery in humans is unknown. Parturition in rats was not affected by olanzapine.
Olanzapine should be used during labor and delivery only if the potential benefit justifies the potential risk.

OVERDOSE

Signs and Symptoms
Very common symptoms in overdose (>10% incidence) include tachycardia, agitation/aggressiveness, dysarthria, various extrapyramidal symptoms, and reduced level of consciousness ranging from sedation to coma.

Other medically significant sequelae of overdose include delirium, convulsion, coma, possible neuroleptic malignant syndrome, respiratory depression, aspiration, hypertension or hypotension, cardiac arrhythmias (<2% of overdose cases), and cardiopulmonary arrest. Fatal outcomes have been reported for acute overdoses as low as 450 mg, but survival has also been reported following acute overdose of approximately 2 g of oral olanzapine.

Management of Overdose

The possibility of multiple drug involvement should be considered. In case of acute overdosage, establish and maintain an airway and ensure adequate oxygenation and ventilation, which may include intubation. Gastric lavage (after intubation, if patient is unconscious) and administration of activated charcoal together with a laxative should be considered. The administration of activated charcoal (1 g) reduced the C_{max} and AUC of oral olanzapine by about 60%. As peak olanzapine levels are not typically obtained until about 6 hours after dosing, charcoal may be a useful treatment for olanzapine overdose.
The possibility of obtundation, seizures, or dystonic reaction of the head and neck following overdose may create a risk of aspiration with induced emesis. Cardiovascular monitoring should commence immediately and should include continuous electrocardiographic monitoring to detect possible arrhythmias.

There is no specific antidote to olanzapine. Therefore, appropriate supportive measures should be initiated. Hypotension and circulatory collapse should be treated with appropriate measures such as intravenous fluids and/or sympathomimetic agents. (Do not use epinephrine, dopamine, or other sympathomimetics with beta-agonist activity, since beta stimulation may worsen hypotension in the setting of olanzapine-induced alpha blockade.) Close medical supervision and monitoring should continue until the patient recovers.

STORAGE: Store below 30ºC, protected from moisture

KEEP ALL MEDICINES OUT OF THE REACH OF CHILDREN

SUPPLY: Blister Pack of 4 x 7's

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